

B.Tech. Degree VI Semester Regular/Supplementary Examination May 2024

ME 19-205-0605 CAD/CAM (2019 Scheme)

Complete answer key including all questions from pages 1-3

PART A

I. (a) Differentiate between constructive solid geometry (CSG) and boundary representation method (BREP).

Answer: Constructive Solid Geometry (CSG) and Boundary Representation (BREP) are two fundamental approaches to solid modeling in CAD systems.

Constructive Solid Geometry (CSG): - Represents objects as combinations of primitive shapes (cubes, cylinders, spheres, etc.) - Uses Boolean operations (union, difference, intersection) to combine primitives - Stores the construction history as a tree structure (CSG tree) - Advantages: Intuitive to use, guarantees valid solids, compact data structure - Disadvantages: Limited in representing complex freeform surfaces, computationally intensive for display

Boundary Representation (BREP): - Represents objects by their bounding surfaces, edges, and vertices - Explicitly stores topological information about faces, edges, and vertices and their relationships - Describes geometry using surface equations and connectivity information - Advantages: Better for complex shapes, efficient for display, supports direct editing - Disadvantages: More complex data structure, may allow invalid solids if not properly implemented

I. (b) What is the production rate and throughput time required for each cycle of an 8-station dial indexing automatic assembly machine powered by a Geneva mechanism operating at a driver rotation speed of 30 revolutions per minute?

Answer: For an 8-station dial indexing automatic assembly machine with a Geneva mechanism:

Given information: - Driver rotation speed = 30 revolutions per minute (rpm)
- Number of stations = 8

For a Geneva mechanism: - Each driver rotation causes one indexing movement - With 8 stations, the dial rotates 45° ($360^\circ/8$) during each indexing

Production rate calculation: - Production rate = Driver speed = 30 units per minute

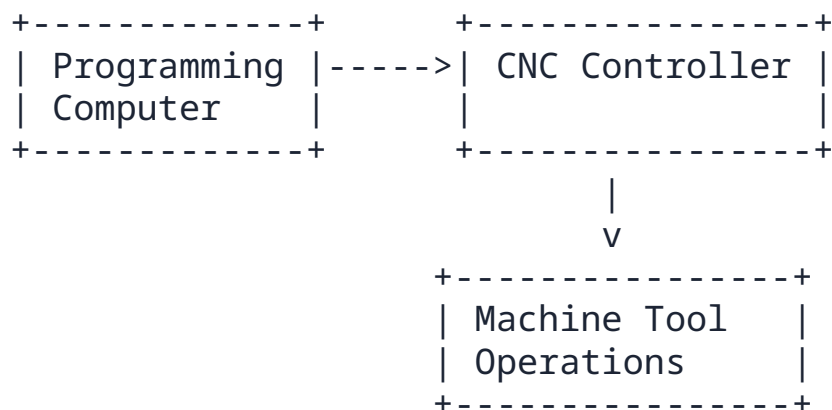
Throughput time calculation: - Throughput time = $1/\text{Production rate} = 1/30$ minute = 2 seconds per cycle

Therefore, the production rate is 30 units per minute, and the throughput time is 2 seconds per cycle.

I. (c) Differentiate between CNC and DNC with neat schematic diagrams.

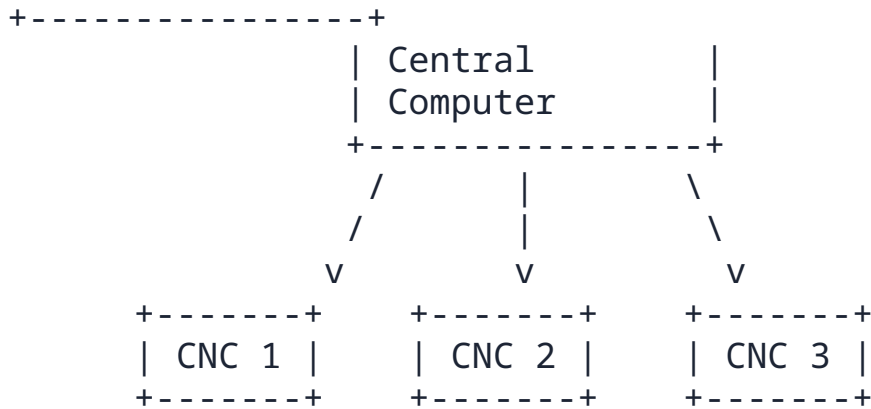
Answer: Computer Numerical Control (CNC): - A standalone machine tool controlled by a dedicated computer - Program stored and executed in the machine's control unit - Limited memory capacity - Typically operates as an independent unit - Programs may be entered manually or transferred via storage media

Schematic for CNC:



Direct Numerical Control (DNC): - A central computer directly controls multiple CNC machines - Programs stored centrally and downloaded to machines as needed - Eliminates need for individual storage at each machine - Allows real-time monitoring and control of multiple machines - Facilitates data collection and production management

Schematic for DNC:



Modern DNC systems have evolved into Distributed Numerical Control, where networked computers manage multiple machines with bidirectional data flow.

I. (d) What are the different types of statements used in APT? Explain each with an example.

Answer: APT (Automatically Programmed Tool) is a high-level language used for programming NC machine tools. It consists of several types of statements:

1. **Geometric Statements:** - Define the geometry of the workpiece - Example: POINT/P1 = (10, 20, 0) - Defines a point P1 at coordinates (10, 20, 0) - Example: LINE/L1 = POINT/P1, POINT/P2 - Defines a line L1 through points P1 and P2
2. **Motion Statements:** - Control tool movement - Example: GOTO/P1 - Moves the tool to point P1 - Example: GOLFT/L1, PAST, P2 - Moves the tool to the left of line L1, past point P2
3. **Postprocessor Statements:** - Provide machine-specific instructions - Example: MACHIN/MILL - Specifies a milling machine - Example: COOLNT/ON - Turns coolant on
4. **Auxiliary Statements:** - Control miscellaneous functions - Example: SPINDL/1000, CLW - Sets spindle speed to 1000 rpm, clockwise direction - Example: FEDRAT/20 - Sets feed rate to 20 units per minute
5. **Pattern Statements:** - Define repetitive operations - Example: COPY/L1, 5, YLARGE, 10 - Copies operation on L1 five times, each 10 units apart in Y direction

6. Computational Statements: - Perform calculations - Example: PARAM/
X = 5 - Assigns value 5 to parameter X - Example: PARAM/Y = X +
3 - Assigns X + 3 to parameter Y

I. (e) Explain forward kinematics and inverse kinematics with respect to robot manipulation.

Answer: Forward Kinematics: Forward kinematics determines the position and orientation of the robot's end-effector given the joint parameters (angles for revolute joints, displacements for prismatic joints). It answers the question: "Where is the end-effector when the joints are at these specific angles?"

The process involves: 1. Establishing coordinate frames for each link using conventions like Denavit-Hartenberg (DH) 2. Computing transformation matrices between consecutive frames 3. Multiplying these matrices to get the final end-effector position and orientation

For example, in a 2-link planar robot with joint angles θ_1 and θ_2 and link lengths l_1 and l_2 : - End-effector x-position = $l_1\cos(\theta_1) + l_2\cos(\theta_1+\theta_2)$ - End-effector y-position = $l_1\sin(\theta_1) + l_2\sin(\theta_1+\theta_2)$

Inverse Kinematics: Inverse kinematics determines the joint parameters needed to place the end-effector at a desired position and orientation. It answers the question: "What joint angles are needed to reach this specific position?"

The process involves: 1. Starting with the desired end-effector position and orientation 2. Solving a set of nonlinear equations to find joint parameters 3. Handling multiple solutions or no solutions (singularities)

For the same 2-link planar robot, given target position (x,y): - $\theta_2 = \pm\arccos((x^2 + y^2 - l_1^2 - l_2^2)/(2l_1l_2))$ - $\theta_1 = \arctan(y/x) - \arctan((l_2\sin(\theta_2))/(l_1 + l_2\cos(\theta_2)))$

Forward kinematics always has a unique solution, while inverse kinematics may have multiple solutions, requiring additional constraints to select the optimal one.

I. (f) Compare the characteristics of PTP (Point-to-Point) and CP (Continuous Path) robots and assign each type of robot to the appropriate applications of loading and unloading, spot welding, arc welding and spray painting.

Answer: Point-to-Point (PTP) Robots: - Move from one specified point to another without controlling the path between points - Focus on endpoint accuracy rather than trajectory - Generally simpler control systems - Higher speeds possible between points - Less expensive than CP robots - Limited in applications requiring controlled paths

Continuous Path (CP) Robots: - Control the entire trajectory between points - Maintain predetermined path and velocity profile - More complex control systems - Capable of following complex curves and contours - Higher precision during motion - More expensive than PTP robots

Application Assignments:

- 1. Loading and Unloading:** - **Appropriate Robot:** PTP - **Rationale:** Only concerned with picking up an object at one point and placing it at another; the path between is not critical
- 2. Spot Welding:** - **Appropriate Robot:** PTP - **Rationale:** Requires precise positioning at weld points but not controlled movement between points; robot needs to move quickly between weld locations
- 3. Arc Welding:** - **Appropriate Robot:** CP - **Rationale:** Requires controlled speed and orientation along the entire weld seam; path and velocity must be precisely controlled for weld quality
- 4. Spray Painting:** - **Appropriate Robot:** CP - **Rationale:** Demands uniform speed and controlled distance from the workpiece along complex contours to ensure even coating and finish quality

I. (g) Explain the basic principles of Rapid Prototyping.

Answer: Rapid Prototyping (RP) refers to a group of techniques used to quickly fabricate physical models and prototype parts directly from 3D CAD data. The basic principles include:

- 1. Layer-by-Layer Construction:** - All RP processes build parts by adding material layer by layer - Each layer represents a cross-section of the 3D model at a specific height - Layer thickness determines resolution and build time

2. **CAD to Physical Part Process Flow:** - Create 3D model in CAD software - Convert model to STL file format (tessellated geometry) - Slice the STL file into thin cross-sectional layers - Build the part layer by layer - Post-process the part (cleaning, curing, finishing)
3. **Material Addition vs. Subtraction:** - Unlike traditional manufacturing (which removes material), RP adds material - Enables complex geometries impossible with conventional methods - Minimizes material waste
4. **Digital Workflow:** - Direct path from digital design to physical object - No need for tooling or molds - Allows quick design iterations
5. **Common RP Technologies:** - Stereolithography (SLA): UV light cures photosensitive resin - Selective Laser Sintering (SLS): Laser fuses powdered materials - Fused Deposition Modeling (FDM): Extrudes thermoplastic material - Digital Light Processing (DLP): Projects light to cure entire layers at once - PolyJet: Jets photopolymer materials that are UV-cured
6. **Applications:** - Concept visualization - Functional testing - Form and fit testing - Manufacturing tooling - Small-batch production

I. (h) What is post processing in additive manufacturing? Why post processing is required?

Answer: Post Processing in Additive Manufacturing: Post processing refers to the various operations performed on a 3D printed part after the printing process is complete to enhance its properties, accuracy, and appearance.

Why Post Processing is Required:

1. **Support Removal:** - Most AM processes require support structures to build overhangs - These supports must be removed mechanically or chemically - Without removal, the part cannot function as intended
2. **Surface Finish Improvement:** - Raw 3D printed parts often have visible layer lines or stair-stepping effects - Post processing techniques like sanding, polishing, or media blasting improve surface quality - Better aesthetics and reduced friction on functional surfaces
3. **Dimensional Accuracy:** - AM parts may have slight dimensional variations - Machining or precision grinding can achieve tighter tolerances - Critical for parts requiring precise fits

4. **Mechanical Properties Enhancement:** - Heat treatment can relieve internal stresses - Sintering or infiltration can increase density and strength - Annealing improves interlayer bonding
5. **Visual Appearance:** - Painting, dyeing, or coating improves aesthetics - UV protection can prevent degradation over time - Brand-specific colors and finishes can be applied
6. **Functionality:** - Threading, tapping, and drilling add functional features - Assembly of multiple printed components - Addition of non-printed components (inserts, electronics)
7. **Sterilization and Cleaning:** - Medical applications require sterilization - Removal of residual uncured resin or powder
8. **Quality Verification:** - Inspection to ensure specifications are met - Non-destructive testing for critical applications

The extent of post-processing depends on the specific AM technology used and the final application requirements of the part.

PART B

II. A 10 station transfer machine is under consideration to produce a component used in the blower. The item is produced by more conventional methods, but demand for the item cannot be met. The manufacturing engineering department has estimated that the ideal cycle time will be $T_c = 1$ min. from similar transfer lines, it is estimated that breakdowns of all types will occur with a frequency of $F = 0.10$ breakdown/cycle and that the average downtime per line stop will be 6 min. The casting I for the component costs ₹114.15 and it will cost ₹76.10 per minute to operate the transfer line. Cutting tools are estimated to cost ₹11.42 per work part. Compute the following measures of line performance: (i) Production rate (ii) Line efficiency (iii) Cost per unit produced.

Answer: Given information: - Number of stations = 10 - Ideal cycle time (T_c) = 1 min - Breakdown frequency (F) = 0.10 breakdown/cycle - Average downtime per line stop = 6 min - Casting cost = ₹114.15 per component - Operation cost = ₹76.10 per minute - Cutting tool cost = ₹11.42 per work part

(i) Production rate:

The actual production time per part includes the ideal cycle time plus the average downtime: - Expected number of breakdowns per cycle = $F = 0.10$ - Expected downtime per cycle = $F \times \text{Average downtime} = 0.10 \times 6 = 0.6 \text{ min}$ - Actual average cycle time = $T_c + \text{Expected downtime} = 1 + 0.6 = 1.6 \text{ min}$ - Production rate = $1/\text{Actual cycle time} = 1/1.6 = 0.625 \text{ parts/min} = 37.5 \text{ parts/hour}$

(ii) Line efficiency:

Line efficiency is the ratio of ideal cycle time to actual cycle time: - Line efficiency = $(T_c/\text{Actual cycle time}) \times 100\% = (1/1.6) \times 100\% = 62.5\%$

(iii) Cost per unit produced:

Cost per unit includes material cost, operation cost, and tooling cost: - Material cost = ₹114.15 per unit - Operation cost = Operation cost rate $\times$ Actual cycle time = ₹76.10 $\times$ 1.6 = ₹121.76 per unit - Tooling cost = ₹11.42 per unit - Total cost per unit = Material + Operation + Tooling = ₹114.15 + ₹121.76 + ₹11.42 = ₹247.33 per unit

Therefore: - Production rate = 0.625 parts/min or 37.5 parts/hour - Line efficiency = 62.5% - Cost per unit produced = ₹247.33

III. (a) An automatic assembly machine with 8 stations, driven by a Geneva mechanism, has an indexing time of 3 seconds and a dwell time of 5 seconds. Under ideal conditions, the machine completes one assembly in 8 seconds, resulting in a production rate of 450 units per hour. Each station malfunctions once every 100 cycles and if any malfunction occurs, the machine jams, requiring an operator to adjust and restart it, which takes 10 minutes. Calculate the following: (i) The percentage of downtime (ii) The new production rate (iii) The line efficiency.

Answer: Given information: - Number of stations = 8 - Indexing time = 3 seconds - Dwell time = 5 seconds - Ideal cycle time = 8 seconds - Ideal production rate = 450 units/hour - Each station malfunctions once every 100 cycles - Machine downtime per malfunction = 10 minutes = 600 seconds

(i) The percentage of downtime:

First, let's calculate the frequency of breakdowns: - Each station malfunctions once every 100 cycles - With 8 stations, the probability of a malfunction in any cycle = $8/100 = 0.08$ - Expected number of cycles between breakdowns = $1/0.08 = 12.5 \text{ cycles}$

For each breakdown: - Production time lost = 10 minutes = 600 seconds -
Ideal cycle time = 8 seconds

Over a long period, such as 100 cycles: - Total ideal production time = $100 \times 8 = 800$ seconds - Expected number of breakdowns in 100 cycles = $100 \times 0.08 = 8$ breakdowns - Total downtime in 100 cycles = $8 \times 600 = 4800$ seconds - Total actual time = $800 + 4800 = 5600$ seconds - Percentage of downtime = $(4800/5600) \times 100\% = 85.71\%$

(ii) The new production rate:

- Ideal production rate = 450 units/hour
- Line efficiency = Ideal time / Actual time = $800/5600 = 0.1429$ or 14.29%
- New production rate = Ideal production rate $\times$ Line efficiency = $450 \times 0.1429 = 64.3$ units/hour

(iii) The line efficiency:

As calculated above, the line efficiency = 14.29%

Therefore: - Percentage of downtime = 85.71% - New production rate = 64.3 units/hour - Line efficiency = 14.29%

III. (b) Calculate the new transformation matrix for a triangle defined by the coordinates A(2,6), B(8,10) and C(10,3), after applying a translation of 4 units in the positive x-direction and 3 units in the positive y-direction.

Answer: To calculate the transformation matrix for the translation, we need to: 1. Express the original triangle vertices in homogeneous coordinates 2. Apply the translation transformation 3. Determine the new coordinates

Step 1: Original triangle in homogeneous coordinates $A(2,6) \rightarrow [2, 6, 1]$
 $B(8,10) \rightarrow [8, 10, 1]$ $C(10,3) \rightarrow [10, 3, 1]$

Step 2: Translation transformation matrix For a translation of $(t_x, t_y) = (4, 3)$, the transformation matrix is:

$$T = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix}$$

$$T = \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}$$

Step 3: Apply transformation to find new coordinates For each point P, the new point $P' = T \times P$

For point A(2,6):

$$A' = T \times A = \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 2 \\ 6 \\ 1 \end{bmatrix} = \begin{bmatrix} 2+4 \\ 6+3 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 9 \\ 1 \end{bmatrix}$$

Therefore, $A' = (6,9)$

For point B(8,10):

$$B' = T \times B = \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 8 \\ 10 \\ 1 \end{bmatrix} = \begin{bmatrix} 8+4 \\ 10+3 \\ 1 \end{bmatrix} = \begin{bmatrix} 12 \\ 13 \\ 1 \end{bmatrix}$$

Therefore, $B' = (12,13)$

For point C(10,3):

$$C' = T \times C = \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 10 \\ 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 10+4 \\ 3+3 \\ 1 \end{bmatrix} = \begin{bmatrix} 14 \\ 6 \\ 1 \end{bmatrix}$$

Therefore, $C' = (14,6)$

The new triangle coordinates are: - A' : (6,9) - B' : (12,13) - C' : (14,6)

IV. (a) What is the difference between absolute and incremental programming?

Answer: Absolute Programming: - Coordinates are referenced from a fixed origin (zero point) of the machine - Each position is specified by its absolute distance from the origin - G-code: G90 mode - Example: Moving to X50 Y30 means moving to a point that is 50 units along X-axis and 30 units along Y-axis from origin - Advantages: - Less prone to error accumulation - Easier to understand and troubleshoot - Easier to modify part dimensions - Disadvantages: - More digits to enter for distant coordinates - Less convenient for repetitive pattern programming

Incremental Programming: - Coordinates are referenced from the current tool position - Each position is specified by its distance and direction from the previous position - G-code: G91 mode - Example: Moving to X20 Y-10 means moving 20 units positive along X-axis and 10 units negative along Y-axis from

current position - Advantages: - Convenient for repetitive patterns and cycles - Often requires fewer digits for coordinate input - Easier to program relative movements - Disadvantages: - Error accumulation can occur over multiple moves - More difficult to troubleshoot - More challenging to modify specific dimensions

Modern CAM systems and CNC controllers typically support both modes, allowing programmers to switch between them as needed for different operations. Many programs use a combination, using absolute mode for critical positioning and incremental mode for repetitive features.

IV. (b) What are G codes and M codes in CNC programming?

Answer: G Codes (Preparatory Functions): G codes are preparatory commands that determine the mode of machine operation and how the motion will be interpreted. They control the type of motion, coordinate system, and other conditions.

Common G codes include: - G00: Rapid positioning (at maximum speed) - G01: Linear interpolation (controlled feed rate) - G02/G03: Circular interpolation (clockwise/counterclockwise) - G17/G18/G19: Plane selection (XY/ZX/YZ) - G20/G21: Inch/Metric unit system - G28: Return to reference point - G40/G41/G42: Cutter compensation (off/left/right) - G43/G44: Tool length compensation (positive/negative) - G54-G59: Work coordinate system selection - G90/G91: Absolute/Incremental positioning - G94/G95: Feed per minute/Feed per revolution

M Codes (Miscellaneous Functions): M codes control auxiliary functions not related to tool motion but to machine operations such as coolant, spindle, tool changes, and program flow.

Common M codes include: - M00: Program stop - M01: Optional stop - M02/M30: End of program - M03/M04/M05: Spindle on clockwise/counterclockwise/off - M06: Tool change - M07/M08/M09: Coolant on mist/flood/off - M10/M11: Clamp/unclamp - M19: Orient spindle - M48/M49: Override enable/disable - M98/M99: Subprogram call/return

The actual implementation of G and M codes may vary slightly between different CNC controllers and manufacturers, though most follow similar standards. Programmers need to refer to the specific machine's programming manual for the exact functionality.

IV. (c) Write a CNC part program to drill 5 holes (R 3 mm) as shown in the figure at an aluminium slab of 19 mm thickness and size 90 mm × 90 mm. Use G81 fixed cycle for drilling.

Answer:

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O1000 (DRILLING 5 HOLES - 3MM RADIUS)
(PROGRAM FOR DRILLING 5 HOLES IN 90X90 ALUMINUM PLATE)
(MATERIAL: ALUMINUM, THICKNESS: 19MM)
(TOOL: 6MM DIAMETER DRILL)

N10 G90 G21 G17 G40 G49 G54 (ABSOLUTE COORDINATES, METRIC, XY PLANE)
N20 G00 Z50.0 (RAPID TO SAFE HEIGHT)
N30 T1 M06 (LOAD 6MM DRILL)
N40 S1200 M03 (SPINDLE ON CLOCKWISE AT 1200 RPM)
N50 G00 X30.0 Y60.0 (RAPID TO FIRST HOLE POSITION)
N60 G43 H01 Z10.0 M08 (TOOL LENGTH COMPENSATION, COOLANT ON)

(DEFINE G81 DRILLING CYCLE)

N70 G81 Z-22.0 R3.0 F100.0 (DRILL CYCLE: DEPTH=-22MM, R-PLANE=3MM,

(DRILL THE 5 HOLES)

N80 X30.0 Y60.0 (FIRST HOLE)
N90 X60.0 Y60.0 (SECOND HOLE)
N100 X45.0 Y45.0 (THIRD HOLE)
N110 X30.0 Y30.0 (FOURTH HOLE)
N120 X60.0 Y30.0 (FIFTH HOLE)

N130 G80 (CANCEL DRILL CYCLE)

N140 G00 Z50.0 M09 (RAPID TO SAFE HEIGHT, COOLANT OFF)

N150 G28 G91 Z0.0 M05 (RETURN TO REFERENCE POINT, SPINDLE OFF)

N160 G28 X0.0 Y0.0 (RETURN TO MACHINE ZERO)

N170 M30 (END OF PROGRAM)

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Explanation: - Lines N10-N20: Set initial conditions (absolute coordinates, metric units, XY plane) - Lines N30-N40: Tool setup and spindle activation - Lines N50-N60: Position to first hole and activate tool length compensation - Line N70: Define G81 drilling cycle with parameters: - Final depth: -22.0 mm (3 mm deeper than the 19 mm thickness for complete through-hole) - R-plane: 3.0 mm (retract height above workpiece) - Feed rate: 100.0 mm/min - Lines N80-N120: Drill the five holes at specified coordinates - Line N130:

Cancel the drilling cycle - Lines N140-N170: Return to safe position, turn off machine functions, and end program

Note: The drill has a 6mm diameter (3mm radius) as specified in the problem. The depth is set to 22mm to ensure through-holes in the 19mm thick material, providing clearance for chip evacuation.

V. Figure shows a two and a half dimensional part. The part thickness is 1 inch. Two holes in the positions shown are to be drilled in the part. Write an APT program to drill the two holes. Use the tool home position shown a drilling speed of 500 r/min. The machine that performs the drilling is number5 and its controller is coded as DRILL.

Answer:

```
PARTNO DRILLING TWO HOLES
MACHIN/5, DRILL
UNITS/INCHES
CLPRNT
$ TOOL DEFINITION
CUTTER/1, 0.75
$ DEFINE POINTS
POINT/P1 = (0,0,0)           $ HOME POSITION
POINT/P2 = (3.25,2.0,0)      $ FIRST HOLE POSITION
POINT/P3 = (7.75,2.0,0)      $ SECOND HOLE POSITION
$ SETUP
SPINDL/500, CLW
COOLNT/FLOOD
LOADTL/1
FEDRAT/3.0
$ DRILLING OPERATION
FROM/P1
GODLTA/0,0,1.0               $ SAFE Z HEIGHT
GOTO/P2
GODLTA/0,0,-2.0              $ APPROACH TO DRILLING POSITION
GODLTA/0,0,-1.0              $ DRILL THROUGH (1 INCH THICKNESS)
GODLTA/0,0,3.0               $ RETRACT
GOTO/P3                      $ MOVE TO SECOND HOLE
GODLTA/0,0,-2.0              $ APPROACH TO DRILLING POSITION
GODLTA/0,0,-1.0              $ DRILL THROUGH (1 INCH THICKNESS)
GODLTA/0,0,3.0               $ RETRACT
GOTO/P1                      $ RETURN TO HOME POSITION
COOLNT/OFF
SPINDL/OFF
```

END
FINI

Explanation: 1. The program starts by defining the part number, machine, and units. 2. The cutter is defined with a 0.75 inch diameter as specified in the drawing. 3. Key points are defined: home position at origin (0,0,0) and the centers of both holes based on the dimensions in the drawing. 4. Machine setup includes setting spindle speed to 500 rpm clockwise, turning on coolant, loading the tool, and setting a feed rate. 5. The drilling sequence: - Move from home position to a safe Z height - Position over the first hole location - Approach and drill through the 1-inch thick part - Retract the drill - Move to the second hole position - Drill the second hole - Retract and return to home position 6. Finally, turn off coolant and spindle, and end the program.

The coordinates are derived from the drawing: - First hole: $X = 2.0 + 1.25 = 3.25$, $Y = 2.0$ - Second hole: $X = 2.0 + 1.25 + 4.5 = 7.75$, $Y = 2.0$

VI. (a) Describe the control loops used in robotic systems. Explain the role of feedback in maintaining the accuracy and stability of robotic movements.

Answer: Control Loops in Robotic Systems:

Robotic systems typically employ several levels of control loops to achieve precise and stable operation. The main control loops include:

- 1. Joint Level Control Loop:** - The most basic control loop that manages individual joint movements - Each robot joint has its own dedicated controller - Typically implements a PID (Proportional-Integral-Derivative) controller - Controls position, velocity, or torque of individual actuators - Operates at high frequency (typically 1-10 kHz)
- 2. Cartesian Level Control Loop:** - Controls the end-effector position and orientation in Cartesian space - Transforms desired Cartesian coordinates into joint coordinates using inverse kinematics - Typically operates at a lower frequency than joint-level control (100-1000 Hz) - May include velocity and acceleration profiles
- 3. Trajectory Planning Control Loop:** - Plans smooth motion paths between waypoints - Generates time-optimal trajectories considering velocity and acceleration limits - Ensures continuity in position, velocity, and acceleration - Operates at a lower frequency (10-100 Hz)

4. **Force/Compliance Control Loop:** - Controls the interaction forces between the robot and its environment - Critical for tasks requiring contact (assembly, grinding, polishing) - May implement impedance control, admittance control, or hybrid position/force control - Often integrated with position control in an outer loop
5. **Visual Servoing Control Loop:** - Uses visual feedback to correct robot position - Closes the loop through machine vision systems - May operate at camera frame rate (30-60 Hz)

Role of Feedback in Maintaining Accuracy and Stability:

Feedback is crucial for robotic systems for several reasons:

1. **Error Compensation:** - Feedback measures the difference between desired and actual states - Controllers use this error to apply corrective actions - Without feedback, errors accumulate and lead to poor accuracy
2. **Disturbance Rejection:** - External forces, payload variations, and mechanical wear affect robot motion - Feedback allows the system to detect and compensate for these disturbances - Maintains performance despite changing conditions
3. **Nonlinearity Compensation:** - Robot dynamics are inherently nonlinear due to coupling between joints, gravity effects, etc. - Feedback helps compensate for these nonlinearities - Enables linear control techniques to work effectively
4. **Stability Enhancement:** - Properly tuned feedback prevents oscillations and overshooting - Dampens the system to ensure smooth motion - Prevents instability that could damage the robot or environment
5. **Adaptation to Parameter Variations:** - Robot parameters (mass, inertia) may change during operation - Feedback allows controllers to adapt to these changes - Maintains consistent performance throughout operation
6. **Performance Verification:** - Feedback provides real-time verification that commands are being executed correctly - Enables fault detection when behavior deviates from expected

The quality of sensors, control algorithm design, and the feedback frequency directly impact the robot's accuracy, repeatability, and stability. Modern robots

employ sophisticated sensor fusion techniques and advanced control algorithms to optimize performance across different operating conditions.

VI. (b) Compare different types of sensors, such as vision systems, range detectors and force sensors. Explain how these sensors enhance robot perception and interaction with the environment.

Answer: Comparison of Different Types of Sensors in Robotics:

Sensor Type	Operating Principle	Key Characteristics	Typical Applications	Limitations
Vision Systems	Process and analyze images from cameras	- High information content	- Part identification	- Sensitive to lighting
		- Non-contact operation	- Quality inspection	- Computationally intensive
Range Detectors	Measure distance to objects	- Can identify objects and features	- Bin picking and tracking	- Occlusion problems
		- 2D or 3D capability	- Visual servoing	- Limited in low contrast
Force Sensors	Measure forces and torques applied	- Direct distance measurement	- Obstacle detection	- Limited field of view
		- Various technologies (laser, ultrasonic, IR)	- Environment mapping	- May have blind spots
		- Medium to high accuracy	- Navigation	- Material-dependent errors
		- Different working ranges	- Proximity sensing	- Interference issues
		- Direct contact measurement	- Assembly operations	- Limited to contact points
		- High precision	- Compliance control	- May be fragile
		- Multi-axis capability	- Quality testing	- Drift over time
		- Fast response time	- Human-robot interaction	- Temperature sensitive

How These Sensors Enhance Robot Perception and Interaction:

1. Vision Systems: - **Enhanced Perception:** - Provide rich contextual information about the environment - Enable object recognition and classification - Allow detection of object positions, orientations, and features - Support quality inspection by detecting defects - **Improved Interaction:** - Guide precise manipulation of objects - Enable adaptive behavior based on visual feedback - Support complex tasks like assembly and sorting - Allow robots to work with unsorted or randomly positioned parts

2. Range Detectors: - **Enhanced Perception:** - Create spatial maps of the environment - Detect obstacles in the robot's path - Measure distances precisely for navigation - Monitor safety zones around the robot - **Improved Interaction:** - Enable collision avoidance - Support autonomous navigation - Allow precise positioning relative to objects - Improve safety when operating near humans or delicate equipment

3. Force Sensors: - **Enhanced Perception:** - Detect contact with objects - Measure interaction forces during manipulation - Sense material properties (stiffness, texture) - Detect assembly completion or part fitting - **Improved Interaction:** - Enable delicate handling of fragile objects - Support compliant motion for insertion tasks - Allow force-controlled processes (polishing, grinding) - Improve safety through contact detection - Enable "teaching" through physical guidance

Integration of Multiple Sensor Types:

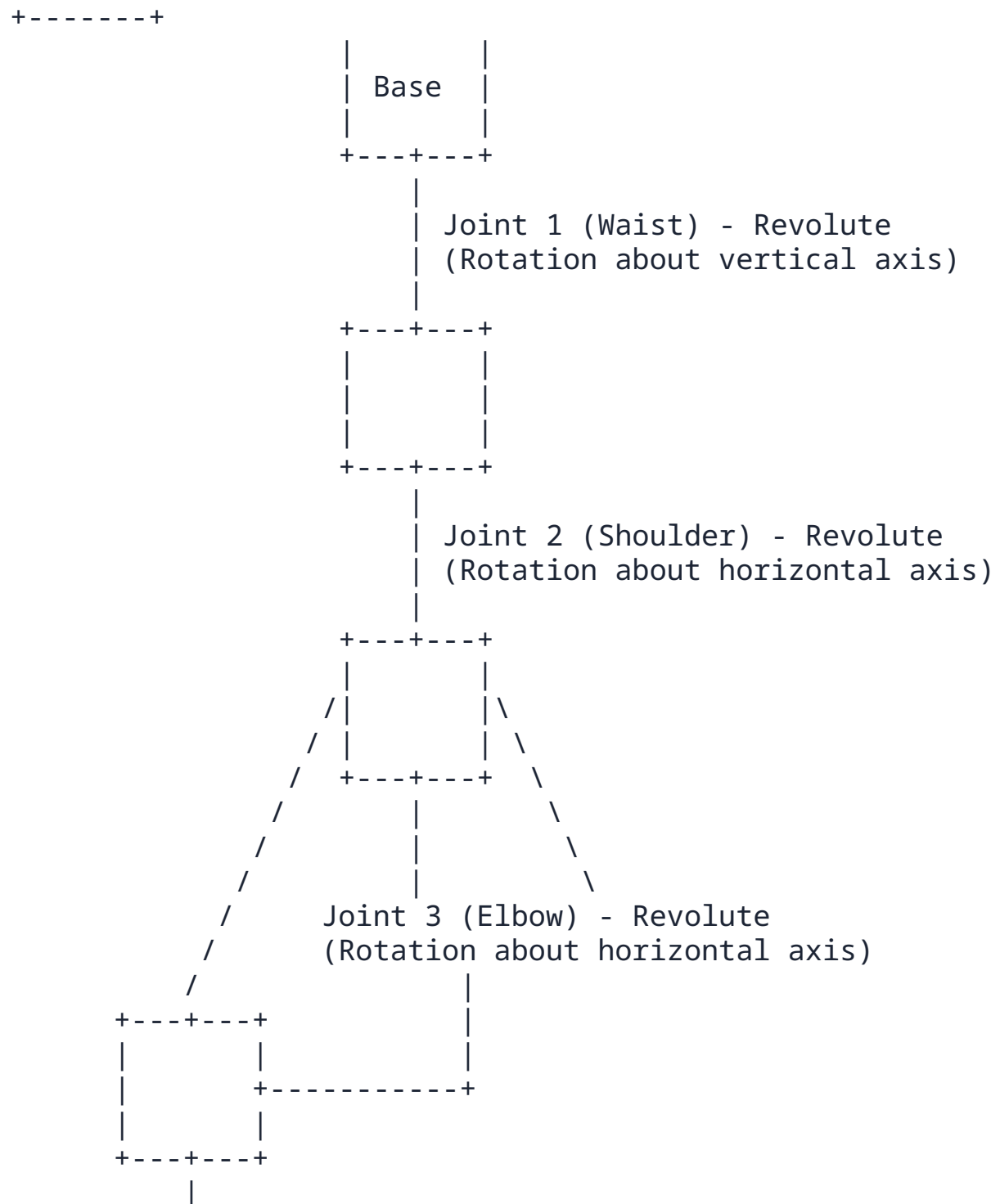
Modern robotic systems often integrate multiple sensor types to overcome individual limitations:

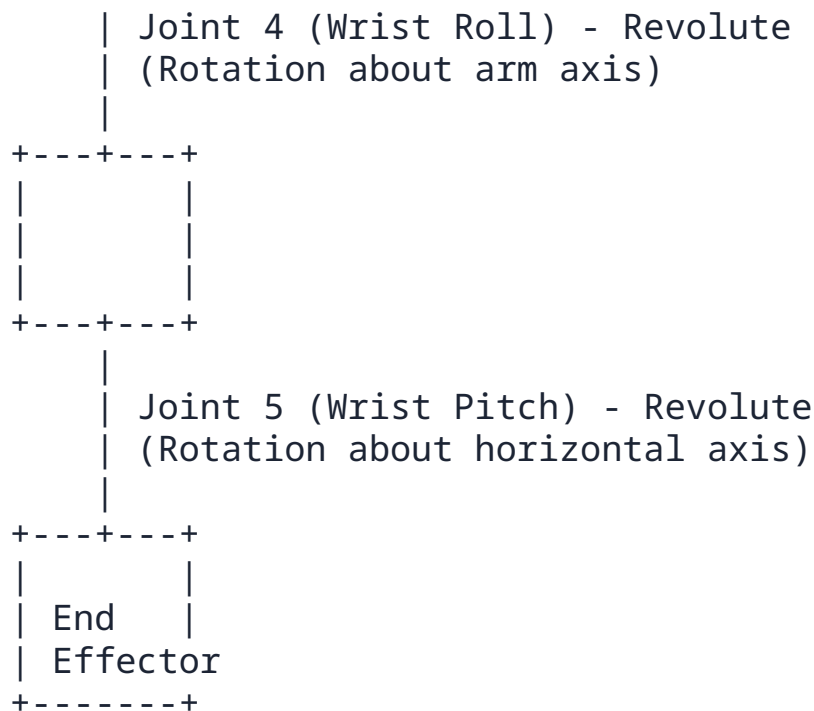
- 1. Sensor Fusion:** - Combining data from different sensors provides more robust perception - Example: Using vision for object identification and force sensing for proper grasping pressure
- 2. Complementary Strengths:** - Vision excels at global awareness and object recognition - Range finding provides precise distance measurements - Force sensing enables controlled physical interaction - Together they create a comprehensive perception system
- 3. Hierarchical Perception:** - Vision for overall scene understanding - Range sensors for intermediate approach and positioning - Force sensors for final interaction and manipulation
- 4. Adaptive Control:** - Multi-sensory feedback enables robots to adapt their behavior based on environmental conditions - Example: Using vision to locate a part, range sensing to approach it safely, and force sensing to handle it appropriately

By integrating these sensor systems, robots can achieve human-like perception capabilities that enable increasingly complex interactions with their environment, adaptive behavior, and safer operation in dynamic settings.

VII. (a) Draw a neat diagram of articulated robot with 5 joints.

Answer:





Key Features of the Articulated Robot: 1. **Joint 1 (Waist)**: Revolute joint allowing rotation about the vertical axis, providing the robot with the ability to reach a wide area around its base. 2. **Joint 2 (Shoulder)**: Revolute joint allowing rotation in the vertical plane, controlling the height and reach of the arm. 3. **Joint 3 (Elbow)**: Revolute joint allowing the forearm to extend or retract, further enhancing the robot's workspace. 4. **Joint 4 (Wrist Roll)**: Revolute joint allowing rotation of the wrist about the arm's longitudinal axis. 5. **Joint 5 (Wrist Pitch)**: Revolute joint allowing the end-effector to tilt up or down.

This 5-joint articulated robot configuration provides: - 5 degrees of freedom (DOF) - Anthropomorphic design resembling a human arm - Versatile workspace with both angular and linear reach capabilities - Ability to approach objects from various angles - Common configuration in industrial applications like welding, assembly, and material handling

Note: This is a 2D representation of a 3D structure. In a real robot, joints 2, 3, and 5 would typically operate in parallel planes, while joints 1 and 4 would allow rotation about perpendicular axes.

VII. (b) Compare and contrast different programming methods used in robotics, such as manual teaching and lead-through teaching and explain the advantages and challenges associated with each method.

Answer:

Comparison of Robot Programming Methods

1. Manual Teaching (Teach Pendant Programming)

Description: Manual teaching involves using a handheld device (teach pendant) to manually guide the robot through desired positions, which are then recorded into the robot's memory.

Process: - Operator uses buttons/joysticks on teach pendant to move robot to desired positions - Each position (waypoint) is saved in the controller's memory - Additional parameters (speed, gripper state, etc.) are specified - Program logic and flow control are added through the pendant interface

Advantages: - Intuitive for operators with minimal programming knowledge - Direct visual feedback during programming - No special software or external equipment required - Immediate verification of robot paths - Suitable for simple pick-and-place tasks - Easy to make minor adjustments on the production floor

Challenges: - Time-consuming for complex paths - Production downtime during programming - Difficult to create optimized paths - Limited capability for complex logic and sensor integration - Repeatability depends on operator skill - Challenging to document and standardize - Difficult to modify without repeating entire sequences

2. Lead-Through Teaching (Direct Teaching)

Description: Lead-through teaching involves physically guiding the robot by hand through the desired motion path, which the system records and can later replay.

Process: - Robot is placed in a freedrive/gravity-compensation mode - Programmer manually moves the robot arm through desired path - System records positions at regular intervals or at specified points - Additional commands and parameters added afterward

Advantages: - Most intuitive programming method - Extremely fast for demonstrating complex paths - Captures natural human movement - No programming knowledge required - Ideal for tasks requiring smooth, continuous motion - Well-suited for arc welding, painting, or sealant application

Challenges: - May lack precision compared to other methods - Limited ability to incorporate complex logic - Difficult to achieve consistent speed

control - May require post-processing to optimize paths - Repeatability may be lower than other methods - Physical limitations for large or heavy robots - Safety concerns during programming

3. Offline Programming (CAD-Based)

Description: Robot programs are created in simulation software using CAD models, without needing access to the physical robot.

Process: - Create virtual environment with robot and workpiece models - Program robot paths using graphical interfaces - Simulate and verify motion - Generate robot-specific code - Download to robot controller

Advantages: - No production downtime during programming - Path optimization before implementation - Collision detection in simulation - Reuse of CAD data from product design - Suitable for complex parts and environments - Easier documentation and standardization - Programs can be tested without risk to hardware

Challenges: - Requires accurate models of robots and environment - Potential discrepancy between simulation and reality - Higher upfront cost for software and training - Calibration challenges between virtual and real world - Requires specialized programming skills - May need fine-tuning on the actual robot

4. Text-Based Programming

Description: Programs are written in the robot manufacturer's specific programming language or in standard languages like Python with robotics libraries.

Process: - Write code in an editor using robot programming language - Define positions, paths, and logic in text format - Compile and load program to robot controller - Test and debug on the physical robot

Advantages: - Precise control over all aspects of robot behavior - Better integration with external systems and sensors - Support for complex logic and conditional operations - Easier version control and documentation - Reusable code components - Scalable for complex applications

Challenges: - Requires programming expertise - Less intuitive than visual methods - Difficult to visualize paths without testing - Potential for syntax errors - Debugging can be time-consuming - Language differences between robot manufacturers

Emerging Methods

Augmented Reality (AR) Programming

- Program robots by visualizing and manipulating virtual paths overlaid on the real environment
- Combines intuitiveness of teach pendant with visualization benefits of offline programming
- Still in development stages but showing promising results

Programming by Demonstration (PbD)

- Advanced version of lead-through teaching using machine learning
- Robot learns not just positions but intent of movements
- Can generalize learned tasks to similar but different situations

Visual Programming

- Block-based or flowchart-style programming interfaces
- Bridging the gap between text-based and teach pendant methods
- Makes complex logic more accessible to non-programmers

Selection Criteria for Programming Methods

The choice of programming method depends on: - Complexity of the task - Available programming expertise - Production volume and variety - Required accuracy and repeatability - Integration with other systems - Available budget - Production downtime constraints

Most modern manufacturing environments use a combination of these methods, selecting the most appropriate approach for each application to balance programming efficiency with performance requirements.

VIII. (a) Explain the process chain in RP in Integrated CAD/CAM environment.

Answer:

Process Chain in Rapid Prototyping (RP) in an Integrated CAD/CAM Environment

The process chain for Rapid Prototyping in an integrated CAD/CAM environment represents a comprehensive workflow from initial concept to final physical prototype. This systematic approach ensures efficient translation of digital designs into tangible objects while maintaining accuracy and functionality.

1. Conceptualization and CAD Modeling

Activities: - Concept development and requirements specification - Creation of 3D solid or surface model using CAD software - Detailed design with dimensional accuracy and feature definition - Design verification for functionality and manufacturability

Tools and Technologies: - 3D CAD systems (SolidWorks, CATIA, NX, Creo, Fusion 360) - Parametric modeling techniques - Feature-based design approaches

Integration Points: - Design databases for component libraries - PDM/PLM systems for version control - Collaborative design platforms

2. CAD Model Optimization for RP

Activities: - Checking and repairing model integrity (eliminating gaps, overlaps) - Verifying wall thickness for manufacturing constraints - Adding support features if needed for overhangs - Optimizing model orientation for build process

Tools and Technologies: - Model healing software - Design for additive manufacturing (DfAM) tools - Model validation utilities

Integration Points: - Direct links between CAD and model preparation software - Automated validation checks - Design guidelines enforcement systems

3. Tessellation and STL File Generation

Activities: - Converting CAD models to tessellated format (typically STL) - Setting appropriate resolution parameters for tessellation - Verifying STL model integrity - Repairing STL data if needed

Tools and Technologies: - STL export utilities within CAD software - Dedicated STL processing software - Mesh manipulation tools

Integration Points: - Direct STL export from CAD environment - Automated quality checking systems - Resolution optimization tools

4. Slicing and Process Parameter Definition

Activities: - Slicing the STL model into horizontal layers - Defining build parameters (layer thickness, build speed) - Generating support structures - Optimizing build path and parameter settings

Tools and Technologies: - Machine-specific slicing software - Build preparation systems - Process simulation tools

Integration Points: - CAM process planning capabilities - Material database integration - Process parameter optimization systems

5. Machine Preparation and Setup

Activities: - Transferring sliced data to RP machine - Material selection and loading - Machine calibration and setup - Build platform preparation

Tools and Technologies: - Machine control software - Material handling systems - Calibration tools

Integration Points: - Network connections to RP equipment - Material tracking systems - Machine monitoring interfaces

6. Part Building (Actual RP Process)

Activities: - Layer-by-layer construction of the physical part - Real-time process monitoring - Quality control during building - Error detection and correction

Tools and Technologies: - Various RP technologies (SLA, SLS, FDM, etc.) - Process monitoring systems - In-process inspection tools

Integration Points: - Real-time feedback to CAM system - Process data collection - Quality monitoring systems

7. Post-Processing

Activities: - Removing support structures - Surface finishing (sanding, polishing) - Heat treatment or curing if required - Painting or coating application

Tools and Technologies: - Support removal tools - Surface treatment equipment - Post-curing chambers

Integration Points: - Post-processing guidelines from CAM system - Finishing requirement specifications - Quality verification systems

8. Inspection and Validation

Activities: - Dimensional verification against CAD model - Functional testing - Surface quality assessment - Documentation of results

Tools and Technologies: - CMM (Coordinate Measuring Machines) - 3D scanners - Optical measurement systems - Functional testing equipment

Integration Points: - Direct comparison to original CAD data - Tolerance analysis tools - Automated inspection reporting

9. Iteration and Refinement

Activities: - Analyzing discrepancies between design and prototype - Implementing design changes in CAD model - Repeating process for refined prototype - Documenting lessons learned

Tools and Technologies: - Design change management systems - Comparison tools - Version control systems

Integration Points: - Feedback to original CAD models - Design history tracking - Knowledge management systems

10. Production Integration (When Applicable)

Activities: - Transitioning from prototype to production - Refining design for manufacturing processes - Creating production tooling based on validated design - Documentation for manufacturing

Tools and Technologies: - Tool design software - Manufacturing planning systems - Production simulation tools

Integration Points: - Direct transfer to production CAM systems - Manufacturing process planning - Production data management

Key Integration Aspects in Modern CAD/CAM Environments

1. **Data Continuity:** Seamless transfer of information throughout the process chain without loss or corruption.
2. **Bi-directional Communication:** Changes at any stage can be communicated back to earlier stages.
3. **Process Knowledge Management:** Capture and reuse of successful process parameters and design guidelines.
4. **Material-Process Relationship:** Integrated databases linking material properties to optimal process parameters.
5. **Unified User Interface:** Common interfaces reducing training needs and improving efficiency.
6. **Concurrent Engineering Support:** Multiple stakeholders working simultaneously on different aspects of the process.
7. **Traceability:** Complete history of the design-to-prototype process for quality assurance and certification.

An effectively integrated CAD/CAM environment for RP significantly reduces time-to-prototype, improves quality, and enhances the overall product development process by creating a digital thread connecting all aspects of design and manufacturing.

VIII. (b) What is the working principle of stereolithography in Rapid Prototyping?

Answer:

Working Principle of Stereolithography (SLA) in Rapid Prototyping

Stereolithography (SLA) was the first commercially available rapid prototyping technology, invented by Charles Hull in 1986. It remains one of the most precise additive manufacturing processes, renowned for its high accuracy and surface finish quality.

Fundamental Concept

Stereolithography is based on the principle of photopolymerization, where liquid photosensitive resin is selectively cured (solidified) by exposure to specific wavelengths of light, typically ultraviolet (UV) laser or LED light sources.

System Components

A typical SLA system consists of:

1. **Vat/Tank:** Contains the liquid photopolymer resin
2. **Build Platform:** The surface on which the part is constructed
3. **Light Source:** Usually a UV laser or LED array
4. **Optical System:** Directs and focuses the light beam
5. **Recoating Mechanism:** Ensures fresh resin covers each new layer
6. **Control System:** Coordinates all components based on the sliced CAD model

Detailed Working Process

1. Pre-Processing

- The 3D CAD model is converted to STL format

- Supporting structures are generated for overhangs and complex features
- The model is sliced into thin horizontal cross-sections (typically 0.05-0.15mm)
- Process parameters are set (layer thickness, cure time, resin type)

2. Build Setup

- The build platform is positioned near the surface of the liquid resin
- Only a small gap equal to the first layer thickness exists between the platform and resin surface

3. Layer Formation Process

In conventional (top-down) SLA: - The UV laser beam, guided by galvanometer mirrors, traces the cross-section pattern on the resin surface - The UV energy triggers a photochemical reaction that solidifies the exposed areas - The platform lowers by one layer thickness - A recoating blade sweeps across the surface to ensure even distribution of fresh resin - The process repeats for each subsequent layer

In bottom-up (inverted) SLA: - The light source (often LED array with DLP) projects through a transparent tank bottom - The build platform is raised by one layer thickness after each layer is cured - Fresh resin flows under the build platform for the next layer

4. Chemical Process During Curing

- Photoinitiators in the resin absorb UV energy
- The photoinitiators generate free radicals
- These free radicals initiate chain polymerization of monomers and oligomers
- Cross-linking occurs, transforming liquid resin into solid polymer
- The degree of cross-linking determines the mechanical properties

5. Post-Processing

- When building is complete, the platform is raised out of the resin tank
- Excess liquid resin is drained back into the vat
- The part is removed from the platform
- The part undergoes cleaning in solvent baths to remove uncured resin
- Post-curing in a UV oven completes the polymerization process
- Support structures are removed manually or with tools

- Surface finishing operations may be performed (sanding, polishing)

Key Technical Parameters

1. **Layer Thickness:** Typically 25-100 microns, affecting resolution and build time
2. **Laser/Light Power:** Determines cure depth and speed
3. **Scan Speed:** How quickly the laser/light source moves across each layer
4. **Hatch Spacing:** Distance between adjacent laser passes
5. **Cure Depth:** How deep the light penetrates into the resin
6. **Resin Viscosity:** Affects recoating process and detail resolution
7. **Build Orientation:** Impacts surface quality and support requirements

Variations of SLA Technology

1. **Digital Light Processing (DLP):** Uses digital micromirror devices to cure entire layers at once
2. **Continuous Liquid Interface Production (CLIP):** Utilizes oxygen-permeable membrane to enable continuous printing
3. **Two-Photon Polymerization (2PP):** Achieves sub-micron resolution through nonlinear optical effects
4. **Masked SLA:** Uses LCD screens as dynamic masks to selectively block light

Advantages of Stereolithography

- High accuracy ($\pm 0.05\text{mm}$ possible)
- Excellent surface finish
- Good detail resolution
- Transparent parts possible
- Wide range of material properties available
- Isotropic mechanical properties

Limitations

- Photopolymers may be brittle
- Material aging and UV sensitivity
- Post-processing requirements
- Limited material selection compared to some other processes
- Higher cost compared to FDM

- Enclosed system requirements for handling liquid resins

Stereolithography continues to evolve with innovations in resin chemistry, light sources, and process control, maintaining its position as a premier technology for high-precision prototyping and increasingly for production applications.

IX. (a) Explain DNC, CAE, CIM, FMS and basic concepts of AI and expert systems for manufacturing automation.

Answer:

Manufacturing Automation Technologies: DNC, CAE, CIM, FMS, AI, and Expert Systems

Direct Numerical Control (DNC)

Definition: Direct Numerical Control (DNC) is a computer system that directly connects to and controls multiple CNC machines simultaneously, enabling centralized program storage, distribution, and management.

Key Components: 1. **Central Computer System:** Stores and manages all part programs 2. **Communication Network:** Connects the central system to machine controllers 3. **Interface Hardware:** Adapters that connect machines to the network 4. **DNC Software:** Manages program transfer and machine monitoring

Functions: - **Program Distribution:** Sends correct part programs to machines as needed - **Program Storage:** Centralized repository for all NC programs - **Program Editing:** Allows modifications without direct machine access - **Data Collection:** Gathers production data, machine status, and performance metrics - **Shop Floor Control:** Coordinates machine operation and scheduling - **Remote Monitoring:** Observes machine status from central location

Evolution: - Originally meant "Direct" Numerical Control (1960s-70s) - Evolved to "Distributed" Numerical Control (1980s-present) - Modern systems use standard networks and may integrate with ERP/MES systems

Benefits: - Eliminates physical media (tapes, cards, etc.) - Reduces program storage limitations at individual machines - Ensures all machines